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# Author: Winfield , date: 2025,
Purpose: Calculate probability distributions.
# Function for binomial distribution
dbinom(4, size=12, prob=0.2)
[1] 0.1328756
# Function for Poisson distribution
ppois(16, lambda=12)
[1] 0.898709
# Uniform distribution
runif(10, min=1, max=3)
[1] 1.017270 2.247037 1.093897 1.479982
[5] 2.915009 1.881633 1.780786 1.095599
[9] 1.884352 2.080578
#Chi-squared Distribution
qchisq(.95, df=7)
[1] 14.06714
#Student t Distribution
qt(c(.025,.975), df=5)
[1] -2.570582 2.570582
#F Distribution
qf(.95, df1=5, df2=2)
[1] 19.29641
#Exponential distribution
pexp(2, rate=1/3)
[1] 0.4865829
#Normal Distribution
pnorm(84, mean=72, sd=15.2, lower.tail=FALSE)
[1] 0.2149176
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